

BIO-4 results at scale factor 1.0

Qwen3 4B FP8, one H100, model startup excluded

	Quail	vLLM baseline	SoL estimate
Query time	29.26 minutes	6.84 hours	14.91 minutes
GPU cost per query	\$1.93	\$27.03	\$0.98
Fresh input tokens	140.6 million	174.6 million	150.8 million
Recomputed KV tokens	18.0 million	50.3 million	0 (assumed)

Quail is 14.04 times faster than the vLLM baseline.